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Eugene Lee

LinkedIn
Google Scholar
Portfolio

Summary

AI Researcher specializing in in-context learning and test-time adaptation for Multimodal Large Language Models. Expertise in model compression, resource-constrained architectures, and multimodal alignment with focus on maximizing data efficiency and continual learning. Proven track record of translating research into production-ready solutions with real-world impact.

Skills: PyTorch, Python, C/C++, CUDA, JavaScript, Node.js, AWS, Docker, iOS, Android, SQL

Education

National Chiao Tung University, Hsinchu, Taiwan *September 2017 - July 2023*
Ph.D. in Electronics Engineering, Advisor: Prof. Chen-Yi Lee; GPA: 4.0/4.0

National Chiao Tung University, Hsinchu, Taiwan *August 2013 - June 2017*
B.S. in Electronics Engineering

Work Experience

University of Cincinnati, Ohio August 2024 – Present
Postdoctoral Researcher, College of Medicine

- Developed novel **visual in-context learning** framework using Large Language Models to assess treatment effects on cancer cells through **few-shot demonstrations**, surpassing state-of-the-art demonstration selection methods while reducing labeled data requirements by 10×.
- Researched **demonstration selection strategies** that improve model reasoning quality and reduce context window size by 40%, directly addressing efficiency challenges in LLM inference.
- Built optimized imaging pipelines with unsupervised learning methods on top of state-of-the-art vision encoders, reducing data processing time by 60% for biological workflows.

Paidge, Inc. January 2024 – October 2025
Founder

- Created an object-intelligence platform combining vision encoders with LLMs for conversational object interaction, deploying **efficient Vision-Language Models (VLMs)** using vLLM for production.
- Designed multi-frame aggregation model with FAISS for **large-scale instance retrieval**, optimizing embedding quality for multi-view retrieval while maintaining sub-100ms query latency.
- Re-engineered lightweight GraphRAG for **high-throughput indexing**, reducing context requirements through structured knowledge representation and achieving 3× faster inference.
- Conducted pilot deployments generating 10+ actionable insights, demonstrating 40% improvement in visitor engagement over traditional audio guides.

National Yang Ming Chiao Tung University August 2023 – July 2024
Postdoctoral Researcher & CEO (Spin-off: Advanced Bio Chips)

- Directed research on **Vision-Language Pipelines** and **efficient attention mechanisms**, reducing visual token count by 60% and achieving 2× speedup in end-to-end latency.
- Led as CEO of university spin-off to commercialize the **BioFPGA platform**, securing ∼$1M in funding and leading cross-functional integration of IC design and bioassays.
- Delivered system architecture optimizations for edge deployment of BioFPGA for on-site Salmonella and E. Coli diagnosis, improving inference throughput by 5× while meeting strict power constraints.

Selected Publications

h-index: 6; Citations: 463; Total: 16

1. Eugene Lee, Yu-Chi Lin and Jiajie Diao, "**Learning to Select Visual In-Context Demonstrations**," In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) Findings, June 2026
2. Eugene Lee, Ting-Yu Chang, Jui-Huang Tsai, Jiajie Diao and Chen-Yi Lee, "**Hierarchical Pre-Training of Vision Encoders with Large Language Models**," In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshops, June 2026
3. Eugene Lee and Chen-Yi Lee, "**NeuralScale: Efficient Scaling of Neurons for Resource-Constrained Deep Neural Networks**," In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), June 2020 (**Oral**)
4. Eugene Lee, Evan Chen and Chen-Yi Lee, "**Meta-rPPG: Remote Heart Rate Estimation Using a Transductive Meta-Learner**," In Proceedings of the European Conference on Computer Vision (**ECCV**), August 2020
5. Eugene Lee, Cheng-Han Huang and Chen-Yi Lee, "**Few-Shot and Continual Learning with Attentive Independent Mechanisms**," In Proceedings of the International Conference on Computer Vision (**ICCV**), October 2021
6. Eugene Lee, Lien-Feng Hsu, Evan Chen and Chen-Yi Lee, "**Cross-Resolution Flow Propagation for Foveated Video Super-Resolution**," In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), January 2023
7. Zheng-Wei Liu, Eugene Lee, Jui-Huang Tsai, Chen-Yi Lee and Jiajie Diao, "**Semantic-Guided Pruning of Visual Tokens for Vision-Language models**," International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**), May 2026
8. Chen, Rui, Eugene Lee, Yuxin Wang, Aditya Yadav, Minling Zhong, Pragti, Yujie Sun, and Jiajie Diao. **A Versatile Near-Infrared Fluorescent Probe for Fast Assessment of Lysosomal Status via a Large Multimodal Model.**, Aggregate: e70118, 2025.
9. Yadav, Aditya, Eugene Lee, Rui Chen, Soryong R. Chae, Yujie Sun, and Jiajie Diao. **Understanding the Role of Polyethylene Glycol Coating in Reducing the Subcellular Toxicity of MXene Nanoparticles Using a Large Multimodal Model.**, Materials Today Bio (2025): 102372.
10. Eugene Lee and Chen-Yi Lee, **PPG-Based Smart Wearable Device with Energy-Efficient Computing for Mobile Health-care Applications**, IEEE Sensors Journal, vol. 21, no. 12, pp. 13564-13573, 15 June, 2021, doi: 10.1109/JSEN.2021.3069460.

Patents, Awards, and Services

Patents: Video display systems (US 12,211,174, 2025); Physiological sensing method (US App 16/441,801).

Awards: Founder University Cohort 11 (2025); Future Tech Awards (2024); Draper University Entrepreneurship (2022); Novatek Ph.D. Fellowship (2020-2022); Broadcom Foundation Scholarship (2019); Synopsys ARC Contest (3rd Place, 2017).

Academic Service:

- **Reviewer:** CVPR ('22 $\sim$ '26), ICCV ('21, '25), ECCV ('22, '24), NeurIPS ('25), WACV ('26).
- **Invited Talks:** "Semiconductor Chips for Fast Medical Testing" (Taiwan-Malaysia Forum '24); "Going Beyond Neural Architecture Search" (Academia Sinica '20); "NeuralScale" (NTU '20).